# Jenny Mai

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Cloud Engineer @ Prolaio

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## Education

**University of Illinois Urbana-Champaign** Class of 2025

*Bachelor of Science in Data Science & Information Sciences with a Computer Science Minor*

*Honors: James Scholar, UIUC iSchool Scholarship, and UIUC International Student Tuition Scholarship*

*University of Melbourne and Nanyang Technological University* *Exchange 2024 & 2025*

## Experience

**Cloud Engineer Intern** Jun 2025 – Present

*Prolaio | Cloud Integration Team* *Chicago, IL*

- Deployed a CI/CD pipeline to automate dependency & microservice updates, cutting manual deployment tasks by 80%.
- Implemented dedicated Kafka sinks for various data objects and engineered real-time broadcast metrics using Micrometer's TimerFactory for granular data streaming monitoring.
- Integrated Renovate across repositories to automate dependency management, enhancing security and service reliability.

**Software Engineer Intern** Jan 2025 – May 2025

*Bumpups | Prompt Engineering Team* *Dallas, TX*

- Designed & optimized prompts using DSPy to improve LLM fine-tuning efficiency & model performance across use cases.
- Developed data preprocessing pipelines and evaluation scripts, achieving 90% alignment with targeted output behaviors.

**Software Engineer Intern** Aug 2023 – Jun 2025

*Illini Media* *Champaign, IL*

- Developed backend systems, optimizing database operations & pipelines to improve publication efficiency by 45%.
- Implemented role-based access controls and security protocols to ensure data integrity and platform security.

**Research Assistant - Project Manager & Fullstack Developer** Jul 2024 – Dec 2024

*C-Lara Lab | Supervisor: Prof. Manny Rayner | Research Paper: shorturl.at/YbhyT* *Melbourne, AU*

- Led a 5-member Agile team to architect a full-stack web application (Django, PostgreSQL, React) for reinforcement learning, utilizing prompt engineering to drive AI Chain of Thought reasoning.
- Engineered a robust CI/CD pipeline for automated deployment to GCP, reducing manual effort by 85%.

**Research Assistant - Data Engineering** Aug 2023 – Present

*UIUC iSchool Research | Supervisors: Prof. Ian Brooks (UIUC), Prof. Yue Guo (UIUC), Prof. Vishala Mishra (Duke)* *Champaign, IL*

- Engineered data integrity pipelines for 25,000+ bilingual patient records using Python & R, conducting statistical analysis to generate novel insights into Pneumococcal Vaccine effectiveness.
- Developed a novel Generative AI tool to automate PEMAT (Patient Education Materials Assessment Tool) evaluations; rigorously validated by scraping archived web data and replicating peer-reviewed studies for performance analysis.

**Data Engineer Intern** Jun 2021 – Mar 2022

*VietAI Research | Supervisors: Thang Luong, Trieu Trinh (Google DeepMind) | research.vietai.org/mtet* *San Francisco, CA*

- Scraped and preprocessed 30,000 medical journals using Selenium, boosting NLP model accuracy by +3.5 BLEU.
- Optimized data pairing with quality controls, enhancing English-Vietnamese translation performance.

## Projects

**Melbourne Real Estate & Prediction** | *Python, Spark, GeoPandas* | GitHub Oct 2024 – Dec 2024

- Developed a predictive real estate investment tool by scraping 12,000 properties with a Python pipeline, enriching them with OpenRouteService APIs, and using Spark to process the data and model housing prices.

**New York Taxi Analysis & Prediction** | *Python, Spark* | GitHub Jul 2024 – Oct 2024

- Engineered a predictive model for NYC taxi fares using Spark that achieved 94% R-squared, leveraging geospatial/temporal analysis and feature engineering to reduce travel costs by 65%.

## Skills

**Tools & Platforms**: Docker, Kubernetes, Google Cloud, CI/CD, Terraform IaC, Git, Apache Kafka, Apache Spark, BigQuery, MySQL, PostgreSQL, Power BI, Unity
**Programming Languages**: Python, Java, Kotlin, C++, SQL, JavaScript, HTML/CSS
**Frameworks & Libraries**: Django, React.js, Node.js, Flask, NumPy, Pandas, Matplotlib, PyTorch